

Samarth Kishor

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Experience

Atalanta Technologies

Formal Methods Engineer

August 2026 – Present

Arlington, VA

Amazon Web Services (Automated Reasoning Group)

Applied Scientist

August 2022 – August 2026

Arlington, VA

- Led the verification of safety properties of a widely-used internal distributed transaction protocol. Used fuzzing and bounded model checking for bug-finding, and SMT solvers for proof.
- Designed and implemented static analyses in Go (program slicing, taint analysis, and syntactic analysis) to verify security properties for the Amazon SSM Agent codebase deployed on millions of hosts.

Amazon Web Services (Region Services)

Software Development Engineer

July 2021 – August 2022

Arlington, VA

- Wrote an end-to-end data transformation compiler in Go for a JSON-like language used to transfer terabytes of customer data per day. Rigorously demonstrated the correctness of the implementation with property-based testing.
- Mentored new hires and interns in Go programming, compilers, and advanced testing techniques.

Amazon Web Services (Region Services)

Software Development Engineer Intern

May 2020 – August 2020

Arlington, VA

- Extended an existing customer-facing AWS service by implementing a new serverless component in Python using Lambda, S3, and Dynamo.
- Service made heavy use of concurrency and parallelism to save customers time and compute resources, and improved the performance of the existing process by over 200%.

Publications

Soundness Checking of Taint Flow Models

Samarth Kishor, Victor Nicolet, Joey Dodds

TBD

Under Submission

- Used LLM agents to generate taint flow models of functions and symbolically checked their soundness.
- The soundness checker uses light-weight static analyses and a Max-SAT solver to discharge proof obligations. Empirically demonstrated that the approach is more efficient than inter-procedural taint analysis in most cases when analyzing Go programs.

Scaling Security Verification to Large Codebases

Linard Arquint, Samarth Kishor, ..., Daniel Kroening, Peter Müller

May 2026

S&P '26: San Francisco, CA

- Applied static analyses to automatically scale the verification of real-world security protocol implementations by enforcing separation logic invariants.
- Implemented the static analyses by extending the first-party Go pointer analysis library.

Education

University of Virginia

B.A. Computer Science, Technology Entrepreneurship Minor

August 2017 – May 2021

Charlottesville, VA

- **TA Experience:** Programming Languages (Head TA), Theory of Computation, Discrete Mathematics
- **Extracurriculars:** Hackathon Club (committee co-chair), Salsa Club (choreographer and performer)

Skills

Programming Languages	<i>Fluent:</i> Go, Python, Java, Javascript · <i>Familiar:</i> OCaml, Rust, Prolog
Human Languages	<i>Fluent:</i> English · <i>Conversational:</i> Spanish, Japanese, Hindi, Urdu
Automated Reasoning	Static analysis, protocol modeling, separation logic, property-based testing
Software Engineering	AWS, distributed systems, NixOS, NoSQL databases, LLM agents